

HAMMER ODDE FYR, Denmark

WMO: 061930

Lat: 55.2982N

Lon: 14.7738E

Elev: 30

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	23.6	25.9	16.1	12.4	25.6	18.9	14.2	27.3	45.4	38.1	41.2	37.8	11.2	120	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	74.6	66.5	72.3	65.7	70.2	64.6	68.3	72.7	66.8	70.9	65.3	69.2	9.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	97.6	71.0	65.0	92.5	69.5	63.6	88.2	68.0	32.6	73.0	31.4	71.0	30.2	69.2	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.6	34.1	30.7	21.1	81.1	3.4	4.5	18.6	84.3	16.6	86.9	14.7	89.4	12.2	92.6		
(5)			19.3	70.6	3.1	3.2	17.0	73.0	15.2	74.9	13.4	76.7	11.1	79.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	35.2	34.8	37.2	43.0	50.7	58.2	63.7	64.4	58.8	51.1	44.0	38.5		
	DBStd	11.49	4.98	4.97	4.54	4.68	4.91	4.11	4.23	3.79	3.84	4.63	4.52	5.08		
	HDD50	2135	458	425	398	217	51	1	0	0	0	42	186	357		
	HDD65	6151	923	845	863	659	445	208	77	58	190	430	631	822		
	CDD50	1541	0	0	0	8	72	247	424	445	264	77	4	0		
	CDD65	82	0	0	0	0	0	4	36	38	4	0	0	0		
	CDH74	117	0	0	0	1	2	10	54	46	4	0	0	0		
	CDH80	15	0	0	0	0	0	1	7	7	0	0	0	0		
(14) Wind	WSAvg	14.6	17.1	16.3	14.9	12.3	11.9	13.1	12.6	13.1	15.0	15.3	15.9	17.4		
(15) Precipitation	PrecAvg	21.2	1.8	1.1	1.3	1.3	1.3	1.6	1.9	1.9	2.2	2.0	2.5	2.2		
	PrecMax	27.2	4.0	3.5	2.6	3.8	4.2	3.5	5.2	7.0	4.8	7.0	4.2	4.4		
	PrecMin	14.8	0.2	0.2	0.2	0.1	0.2	0.3	0.2	0.1	0.2	0.2	0.7	0.5		
	PrecStd	3.1	0.8	0.7	0.7	0.9	0.8	0.9	1.3	1.6	1.1	1.5	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	49.2	52.6	63.8	68.4	74.9	78.7	79.1	73.4	64.6	55.5	49.9		
		MCWB	45.0	44.7	45.8	51.6	58.2	63.9	69.0	68.2	64.1	56.9	52.5	47.7		
	2%	DB	44.5	45.0	47.3	56.6	63.6	69.7	74.7	74.2	68.9	60.8	53.0	47.7		
		MCWB	42.7	42.1	43.5	49.5	56.7	62.6	67.3	66.7	62.7	56.9	51.1	46.1		
	5%	DB	43.0	42.4	44.8	52.4	60.3	66.9	72.5	72.0	66.2	58.8	51.3	46.2		
		MCWB	41.5	40.5	42.0	47.1	54.7	60.8	66.3	65.9	61.1	55.8	49.3	44.8		
	10%	DB	41.6	41.0	42.7	49.4	57.7	64.6	70.3	70.2	64.1	57.2	49.8	44.8		
		MCWB	40.3	39.6	40.4	45.5	53.0	59.6	65.1	64.8	59.9	54.7	47.9	43.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	45.5	45.1	47.0	53.7	60.8	66.1	72.0	70.1	66.2	59.4	53.5	48.5		
		MCDB	46.7	47.8	50.4	58.7	65.9	71.9	75.3	75.6	70.4	61.8	54.5	49.1		
	2%	WB	43.1	42.6	44.1	50.2	57.7	63.5	68.9	68.3	63.5	57.5	51.4	46.6		
		MCDB	44.2	44.2	46.5	55.4	62.0	68.2	73.0	72.7	67.1	59.9	52.7	47.6		
	5%	WB	41.8	40.9	42.2	47.9	55.6	61.7	67.1	66.9	61.8	56.3	49.7	45.0		
		MCDB	42.8	42.1	44.3	51.8	59.4	65.7	71.3	70.9	65.0	58.5	51.1	46.0		
	10%	WB	40.5	39.6	40.7	45.8	53.7	60.2	65.6	65.5	60.5	54.9	48.1	43.6		
		MCDB	41.4	40.8	42.3	48.8	57.1	63.9	69.6	69.2	63.4	56.9	49.5	44.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.5	4.7	5.8	7.8	8.6	8.3	7.9	7.3	6.0	5.2	4.5	4.5		
		MCDBR	4.8	6.5	9.3	13.6	13.6	12.1	11.4	10.1	9.1	7.2	5.5	4.7		
	5% WB	MCWBR	4.7	4.9	6.6	8.2	8.3	7.6	6.7	6.4	6.0	5.3	4.8	4.8		
		MCWBR	4.7	6.0	8.6	12.5	12.4	10.8	10.2	8.8	7.7	6.5	5.3	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.321	0.344	0.367	0.400	0.397	0.389	0.413	0.415	0.381	0.365	0.346	0.314		
	taud		2.299	2.316	2.303	2.279	2.355	2.408	2.360	2.356	2.424	2.386	2.322	2.287		
	Ebn at Noon		189	229	252	260	268	272	262	253	247	219	176	162		
	Edh at Noon		18	25	31	37	36	35	36	34	28	23	17	15		
(44) All-Sky Solar Radiation	RadAvg		132	330	777	1375	1805	1938	1764	1421	1001	476	184	94		
	RadStd		24	60	71	134	174	117	191	106	97	77	32	12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	-184	N/A	N/A
(47) Regional (51 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	-277	+108	N/A

Nomenclature: See separate page